Step 3h: wedge and positional contests, a short comparison

Four variants, built as engine settings and not adopted. Short budget: one complete solver level per position (10 s budget), 2 seeds; play-outs at the computer's 0.5 s budget.

Findings. None of the three new variants does what the change is for. Contest foul rolls stay rare in every variant: none in A, 2 in 34 possessions in B and C, 1 in D (and-ones 0 or 1 in each). The reason is the reaction: a defender who can still react uses it to reach a Front contest (59% to 67% of shots under B, C and D), and one who can't reach the shooter's line now cannot contest at all, so more shots go up with no eligible contester (21% in A, 26% to 34% in the others). Interceptions roughly double in every new variant, including D with a single arrow (won per possession 0.26 in A, 0.56 to 0.65 in B, C and D): the contest geometry changes how both computers use their arrows, not only the wedge. Card values move little (means 22.7 to 24.1 against 23.4), but card verdicts change on 5 (D), 8 (C) and 9 (B) of the 16 cards at this budget, with both seeds agreeing in every variant. Reactions spent turning on the spot were already rare (5% in A) and disappear in the others. The plan change rate is about the same everywhere (72% to 81%). The play-outs are small (34 possessions a variant), so read the per-possession rates as directions, not as exact figures.

Recommendation. Ranked by the hierarchy. 1. Keep A, the current rules (System Fidelity): no variant makes the foul roll or the and-one appear, and each shifts verdicts on a third to over half of the cards and doubles interceptions, a large change with no gain yet shown. A wrong rule is worse than a slow one. 2. If positional contests are still wanted, test D next, not the wedge (Legibility): it changes the fewest verdicts, keeps one arrow, and B and C show the wedge adds nothing D lacks. 3. The lever that would make Side and Behind appear is the reaction's freedom to turn: a variant where a reaction Move keeps the defender's facing (a Quick Step already does) would test that directly, at some cost to the agency the proposal keeps (Player Agency).

1. Who contests, and from where

Shots in the play-outs (All-Star v All-Star, 2 seeds from each of the 16 cards and the golden position). Under A the category is descriptive (by rim distance, the measure the current −3 uses); there a contest is "clean" or a foul roll by facing.

Variant	Shots	Front	Side	Behind	No eligible contester	Contest declined
A: single arrow, current	28	32%	43%	0%	21%	4%
B: wedge, wedge_mirror	27	67%	4%	4%	26%	0%
C: wedge, rim_distance	29	59%	0%	7%	34%	0%
D: single arrow, rim_distance	28	64%	0%	4%	32%	0%

2. Foul rolls and and-ones

Variant	Contest foul rolls / possession	Fouls given / possession	And-ones / possession
A: single arrow, current	0.00	0.00	0.03
B: wedge, wedge_mirror	0.06	0.00	0.03
C: wedge, rim_distance	0.06	0.03	0.03
D: single arrow, rim_distance	0.03	0.00	0.00

3. Passing lanes and drives

A Dribble is how a ball carrier crosses an adjacent defender's arrow line, so Dribbles per possession is the measure of drives the lines force.

Variant	Interceptions tried / possession	Interceptions won / possession	Dribbles / possession
A: single arrow, current	0.56	0.26	0.85
B: wedge, wedge_mirror	0.91	0.56	1.32
C: wedge, rim_distance	1.09	0.65	0.74
D: single arrow, rim_distance	0.94	0.59	1.56

4. Possession value and bands

Solver values: the 16 cards at a short budget (one complete search level, seed 0) and the golden position (seeds 0 and 1). Play-out values are the points scored.

Variant	Cards: mean value	Golden (2 seeds)	Play-outs: mean points	Cards Excellent	Cards Great	Cards Good	Cards Okay	Cards Poor
A: single arrow, current	23.4	33.0 / 33.0	22.9	2	3	8	2	1
B: wedge, wedge_mirror	22.7	29.8 / 29.8	21.5	3	3	7	1	2
C: wedge, rim_distance	23.1	29.8 / 29.8	21.4	4	3	6	1	2

Variant	Cards: mean value	Golden (2 seeds)	Play-outs: mean points	Cards Excellent	Cards Great	Cards Good	Cards Okay	Cards Poor
D: single arrow, rim_distance	24.1	33.0 / 33.0	24.4	2	3	10	0	1

5. Plan change rate (the context test)

Each card re-solved with the two other matchups round 2 used; the rate is how often the best first play changes. Lower means the card's plan depends less on who is playing.

Variant	Plan change rate	Pairs
A: single arrow, current	81%	32
B: wedge, wedge_mirror	81%	32
C: wedge, rim_distance	75%	32
D: single arrow, rim_distance	72%	32

6. How shot reactions were spent

Variant	Reaction actions	Turning on the spot	Moving	Other (steal)
A: single arrow, current	22	5%	73%	23%
B: wedge, wedge_mirror	21	0%	86%	14%
C: wedge, rim_distance	19	0%	74%	26%
D: single arrow, rim_distance	18	0%	78%	22%

7. Card verdicts that change from A

Card	A	B: wedge, wedge_mirror	C: wedge, rim_distance	D: single arrow, rim_distance
1	Gift	Gift	Gift	Gift
2	Puzzle	Wall *	Wall *	Single trick *
3	Single trick	Puzzle *	Gift *	Puzzle *
4	Wall	Puzzle *	Puzzle *	Wall
5	Gift	Single trick *	Single trick *	Single trick *
6	Gift	Gift	Gift	Gift
7	Gift	Gift	Gift	Gift
8	Wall	Single trick *	Single trick *	Single trick *
9	Single trick	Puzzle *	Puzzle *	Single trick
10	Puzzle	Single trick *	Single trick *	Single trick *
11	Puzzle	Puzzle	Puzzle	Puzzle
12	Puzzle	Puzzle	Puzzle	Puzzle
13	Single trick	Puzzle *	Puzzle *	Single trick
14	Single trick	Single trick	Single trick	Single trick
15	Puzzle	Single trick *	Puzzle	Puzzle
16	Single trick	Single trick	Single trick	Single trick

* differs from A. At this budget the two seeds agree on 16/16 (A), 16/16 (B), 16/16 (C), 16/16 (D) verdicts.

Notes

Variant A's Front / Side / Behind shares are descriptive (rim distance), since under the current rules every adjacent defender contests and position only sets the -3. "Contest declined" means the defence chose not to contest although a defender could. A verdict at this budget comes from one complete solver level (the round-2 verdicts used 90 s and 10 min budgets), so the A verdicts here are not the Solver Report verdicts. Grid questions for the designer are in OPEN_QUESTIONS 59 to 62.